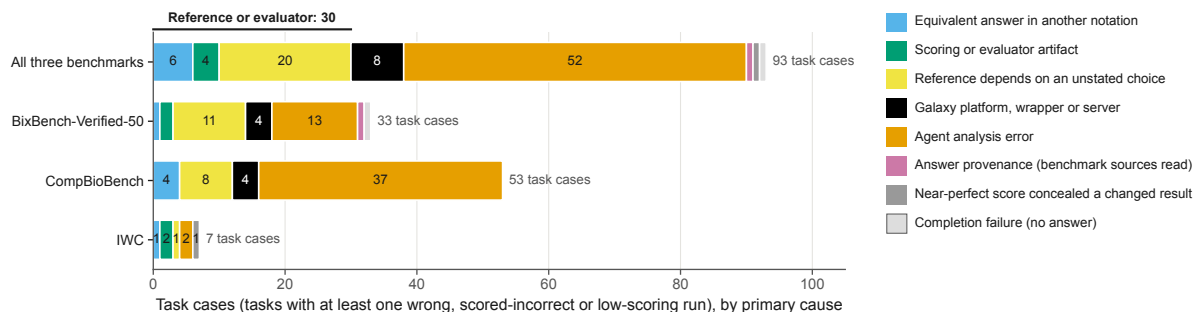


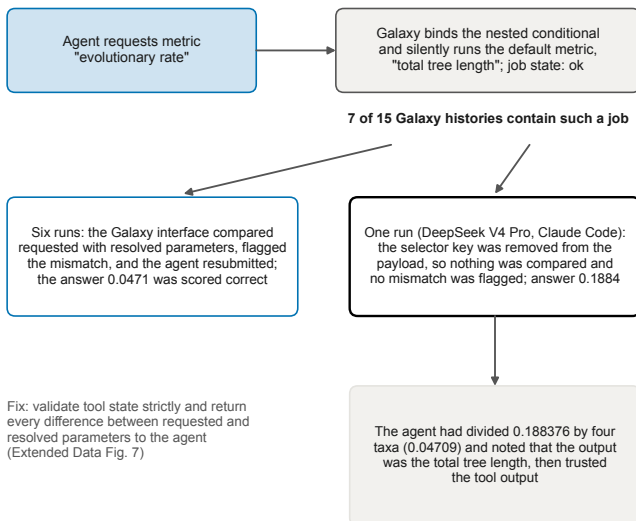
a In 30 of 93 audited task cases the primary cause was the reference or evaluator, not the agent or Galaxy



Recorded in addition to the primary cause: Galaxy contributed as primary or secondary cause in 14 task cases; benchmark answers were retrieved, or retrieval was attempted, in 26; a Galaxy-condition answer was computed in the local shell in 12 (15 correct CompBioBench answers); an answer was copied from another run through the shared Galaxy account in 2 (Extended Data Fig. 8).

b bix-35-q1: an interface-binding failure that returned a successful job

Task: PhyKIT evolutionary rate of one BUSCO gene tree; reference 0.0471.
Scored correct: open-ended code 15 of 15, Galaxy 14 of 15 runs.



c contaminated-rna-q1: a reference database the workbench did not reliably provide

Task: name the contaminant species in a mostly human RNA-seq file (96,053 reads); reference answer *Hydra vulgaris*.
Scored correct: open-ended code 12 of 13 (GPT-6 Astra included), Galaxy 9 of 12 runs.

